



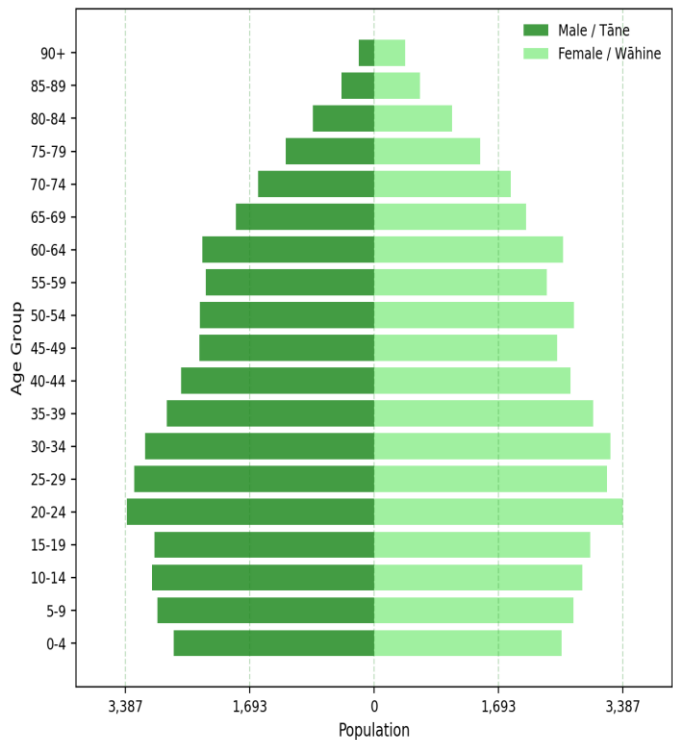
Community Wellbeing Profile

Population and Area Summary

In **Palmerston North City**, the total population in 2023 was **87,099**, covering an area of **394.74 km²**. This gives a population density of **220.6** people per **km²** across **41** Statistical Area 2 (SA2) zones. The population structure shows that **32.42%** of residents are young people (aged under 20), **51.83%** are of working age (20-64 years), and **15.76%** are elderly (65 years and older).

Wards in this TA are: Te Hirawanui General Ward.

The population pyramid shows the age and gender structure of people in **Palmerston North City**. The bars on the left show males (Tāne) and the bars on the right show females (Wāhine). A wide base means there are more young people, while a narrow base shows fewer children and an older population. This helps to understand the balance between young and old people living in Palmerston North City.



Total Population: 87,099
Year: 2023

1 Crime Overview

This section summarises reported victimisations within **Palmerston North City** for the period **2022-2024**, grouped by key ANZSOC offence categories. The Figure 1.1 presents a bar chart, where each row represents a offence type and each bar shows the total number of victimisations across years. Darker green shades indicate more recent years, making it easy to compare trends across time. This visualisation provides a clear overview of how different crime types contribute to the overall volume of reported incidents and how these patterns have changed over time.

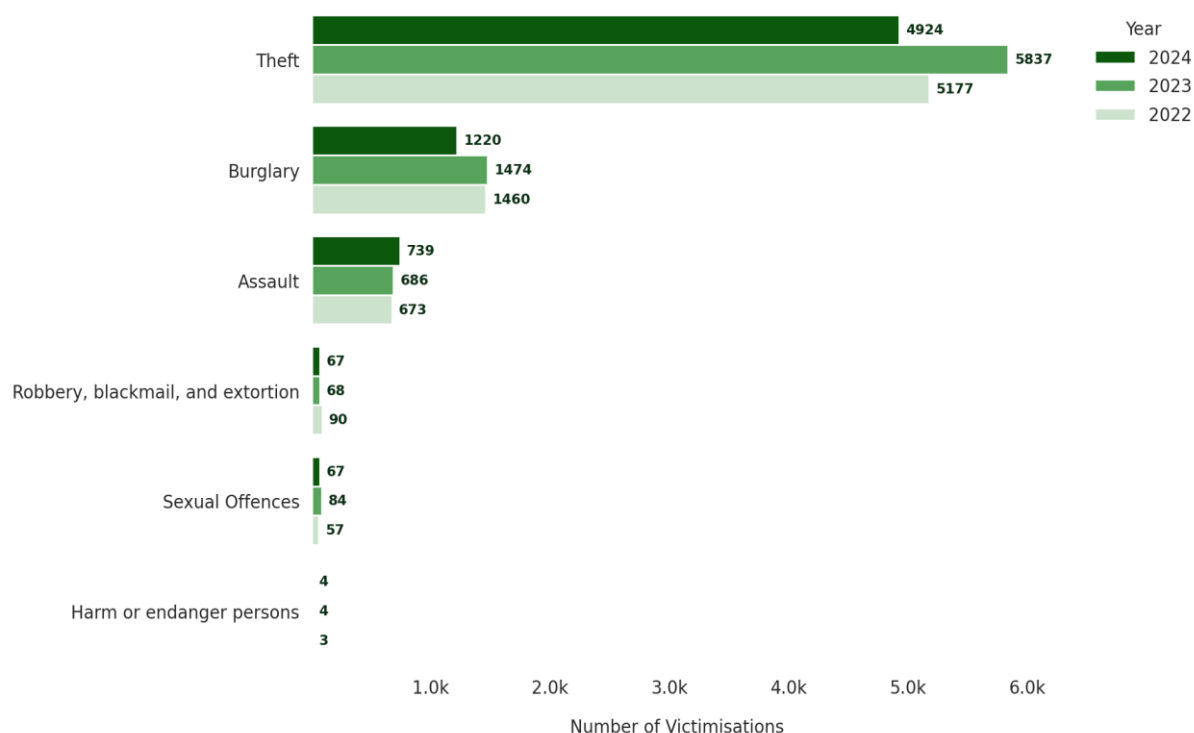


Figure 1.1: Reported victimisations by offence type in Palmerston North City, 2022-2024.

Figure 1.2 compares the total population of each Territorial Authority (TA) with its overall victimisation **rate per 1,000 people** in the year **2023**. Each bubble represents one TA, where the bubble size corresponds to the total number of reported victimisations. **Palmerston North City** is highlighted as the dark green bubble to show its position relative to other Territorial Authorities. Its placement helps to visualise how **Palmerston North City** compares in both total population and victimisation rate. Bubbles positioned higher on the chart indicate areas with higher **victimisation rates**, while those further to the right represent larger **populations**. This visual helps to identify whether higher population centres experience proportionally more or fewer reported offences relative to smaller communities, and where **Palmerston North City** sits within the national distribution. The total recorded crime rate was **93.6 victimisations per 1,000 people**.

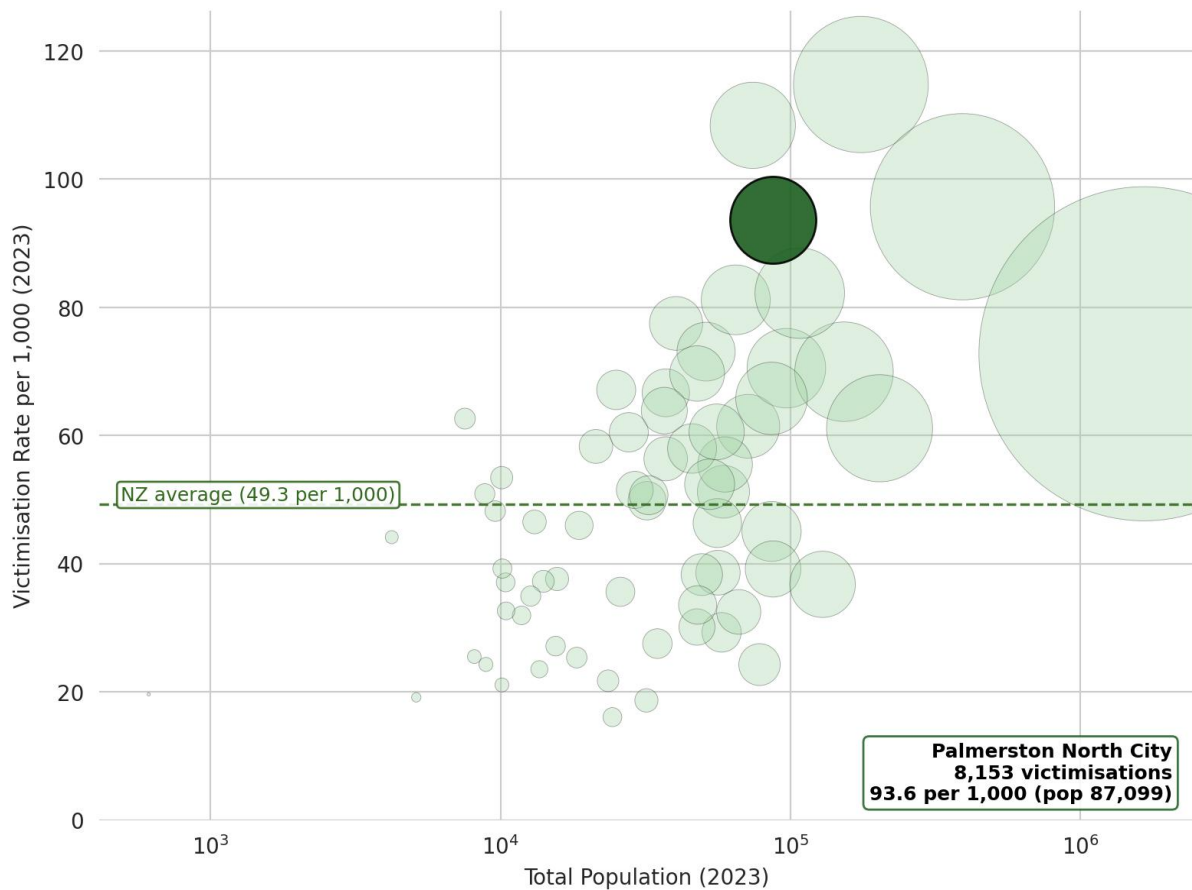


Figure 1.2: Population vs victimisation rate across TAs.

2 Environment Overview

Figure 2.1 provides an overview of key environmental conditions across New Zealand for **2023**. The three maps show average temperature, total rainfall, and greenness (NDVI). Darker green areas indicate higher values, while lighter areas indicate lower values. These national maps help place **Palmerston North City** within a wider context by showing how warm, wet, and green different parts of the country were during the year.

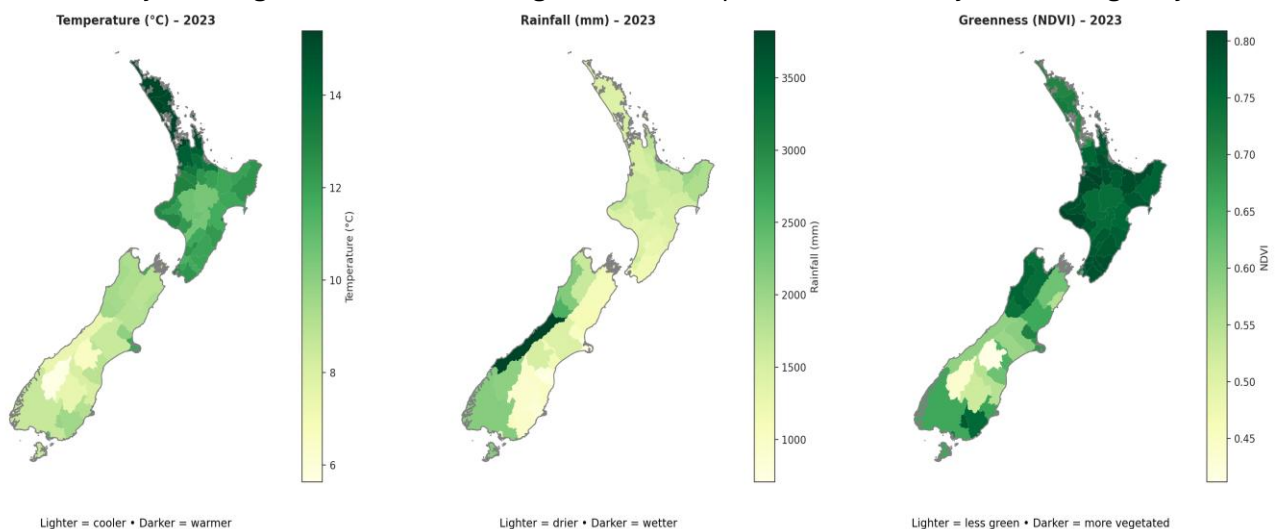


Figure 2.1: National Temperature, Rainfall, and Greenness in 2023

Figure 2.2 shows the spatial variation of temperature, rainfall, and vegetation greenness (NDVI) across the Statistical Areas (SA2s) inside Palmerston North City for 2023. Darker green areas represent relatively warmer,

wetter, or more vegetated locations, while lighter areas indicate cooler, drier, or less vegetated zones. These patterns often reflect local topography, elevation, coastlines, and land cover. Together, the maps help identify how climate and vegetation differ within the TA, showing both broad patterns and localised environmental differences.

Note: Areas shown in light grey represent Statistical Areas (SA2s) where environmental data were unavailable. These typically occur in coastal, harbour, or water-dominated regions where satellite observations (ERA5 or MODIS) could not provide reliable temperature, rainfall, or vegetation values. Because ERA5-Land has ~9 km resolution, some SA2 polygons (especially tiny coastal ones) do not intersect a pixel.

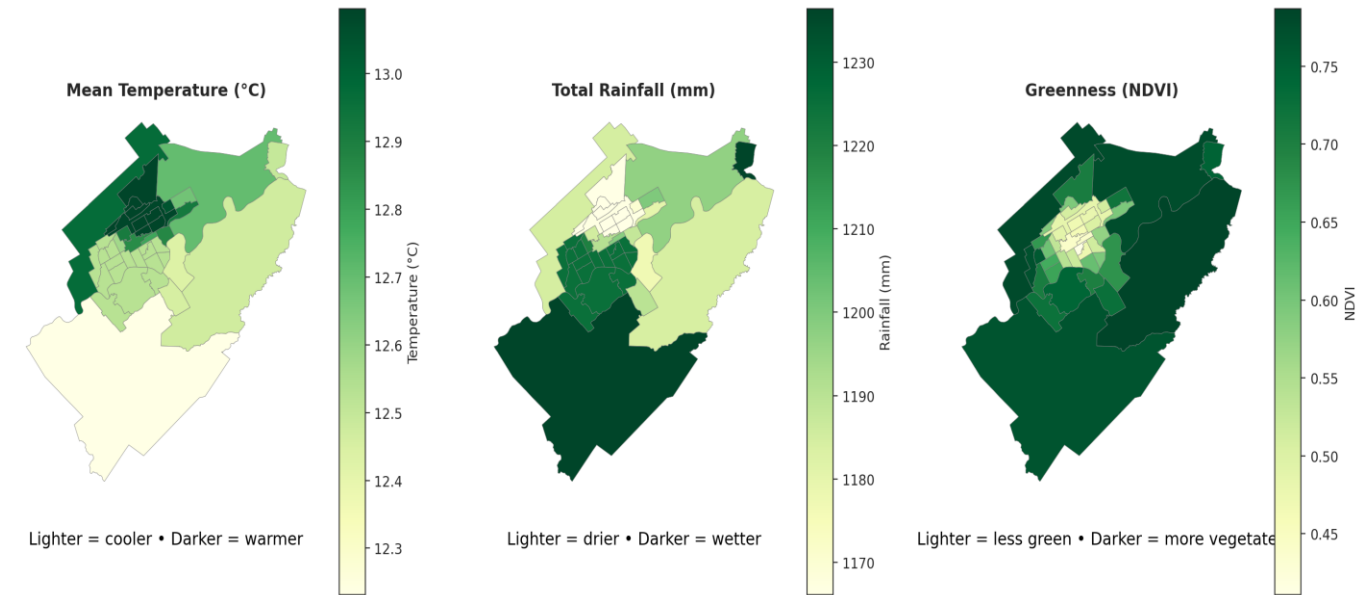


Figure 2.2: Temperature, rainfall, and greenness (NDVI) variation within Palmerston North City.

Table 2.1 compares key environmental indicators for **Palmerston North City** with national averages. The table shows the TA’s mean values, the range across its SA2s, and how these compare to New Zealand overall.

Indicators	TA Mean	TA Range	NZ Mean	NZ Range	TA vs NZ Avg
Mean Temperature (°C)	12.71	12.23-13.1	13.28	4.93-16.91	Same
Total Rainfall (mm)	1,201	1,166-1,236	1,350	656-4,122	Lower
Greenness (NDVI)	0.61	0.41-0.79	0.61	0.12-0.83	Same

Table 2.1: Environmental Indicators - Palmerston North City vs National Averages

Figure 2.3 shows average annual **PM_{2.5}** and **NO₂** concentrations across all New Zealand Territorial Authorities for the period **2018-2022**. The top panels show PM_{2.5} (fine particles), and the bottom panels show NO₂ (nitrogen dioxide). Darker shades represent higher concentrations.

Note: PM_{2.5} and NO₂ data taken from satellites. The NO₂ values show how much nitrogen dioxide is in the air above each area, not the amount at ground level. All data were averaged for the year and matched to each TA using Google Earth Engine. PM_{2.5} means very small dust and smoke particles in the air. These tiny particles can go deep into the lungs. They mainly come from car exhaust, wood burners, industrial smoke, dust, and fires. PM_{2.5} is measured as solid particles in the air (µg/m³). NO₂ is a gas, not a particle. It mostly comes from vehicles, trucks, buses, factories, and other fuel-burning activities. In this report, NO₂ is shown as the amount of gas in the air column above each area, measured in mol/m². The values are very small (about 10⁻⁶ mol), so they appear in scientific notation.

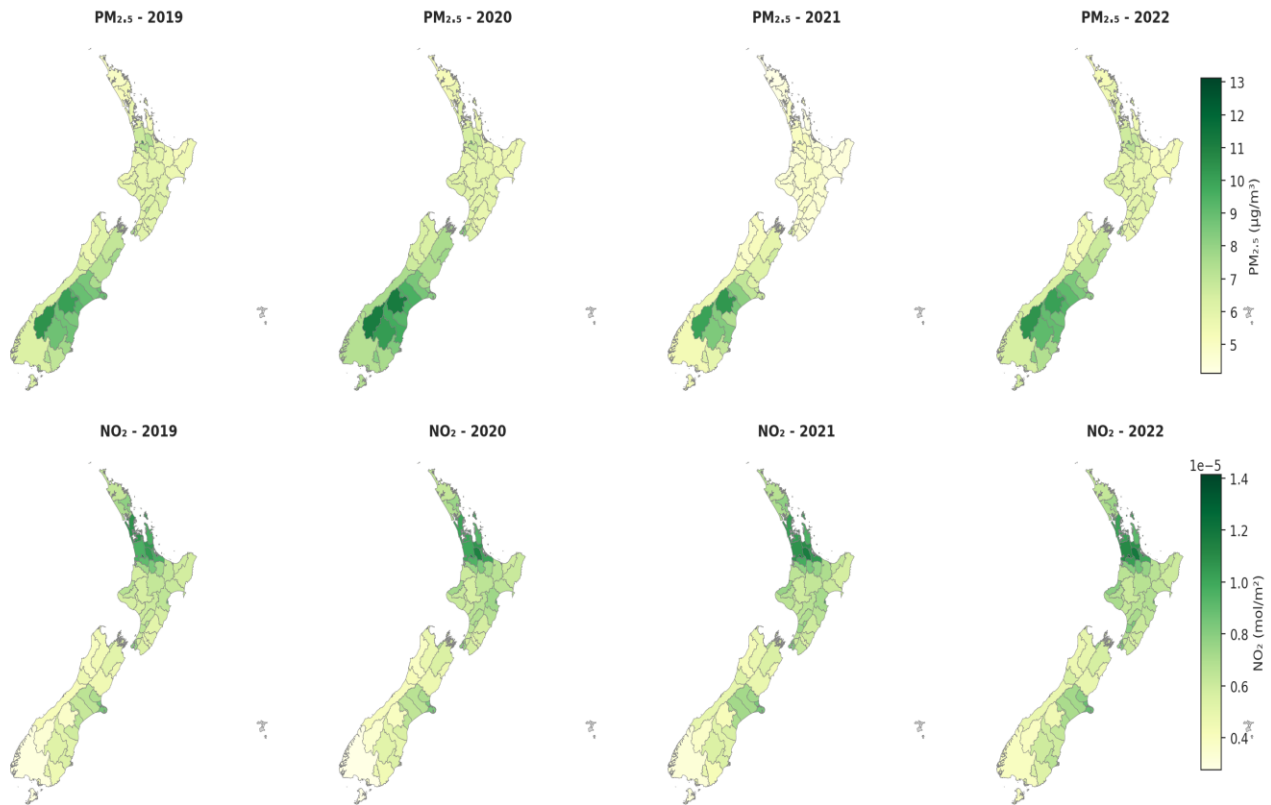


Figure 2.3: TA-level air quality ($PM_{2.5}$ and NO_2).

Figure 2.4 presents the annual $PM_{2.5}$ and NO_2 levels for **Palmerston North City** across **2018-2022**. Bars display the measured value for each year, and the line shows the overall trend. This helps identify whether air quality is improving, worsening, or remaining stable over time.

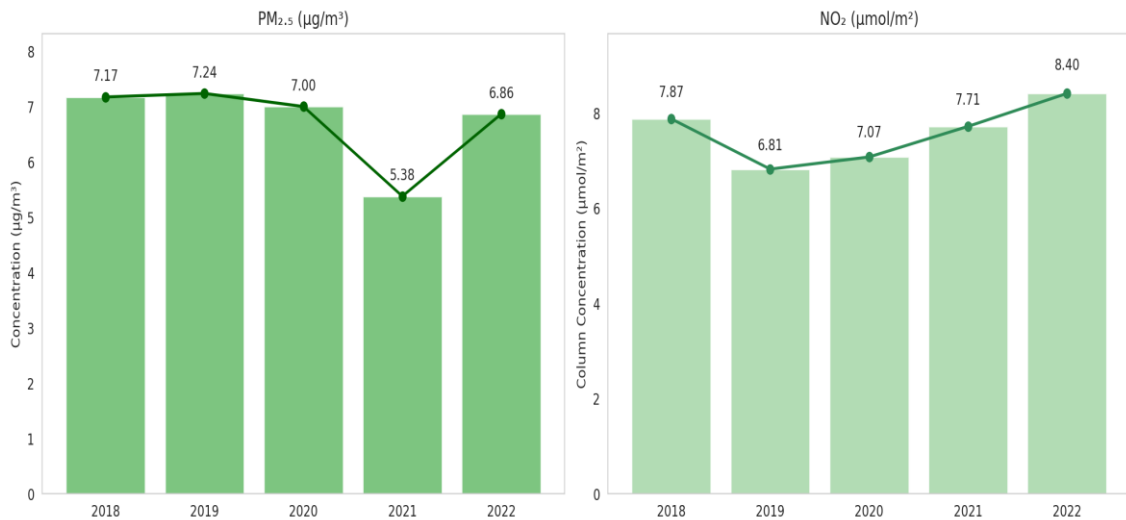


Figure 2.4: Annual $PM_{2.5}$ and NO_2 Trends in Palmerston North City

Table 2.2 compares annual NO_2 ($\mu mol/m^2$) levels in Palmerston North City with national averages. The table shows the TA's mean concentration for each year, the national mean, the national range across all TAs, and whether Palmerston North City is higher or lower than the NZ average.

Indicator (Year)	TA Mean	NZ Mean	NZ Range	TA vs NZ Avg
NO_2 (2018)	7.9	6.9	3.0-13.3	Higher
NO_2 (2019)	6.8	6.5	3.0-12.3	Higher
NO_2 (2020)	7.1	6.7	2.8-12.9	Higher
NO_2 (2021)	7.7	7.0	3.2-13.2	Higher
NO_2 (2022)	8.4	7.1	3.9-14.1	Higher

Table 2.2: Air Quality Indicator NO₂ (µmol/m²) - Palmerston North City vs National Averages

Table 2.3 summarises PM_{2.5} (µg/m³) levels for Palmerston North City compared with national averages. It presents TA mean values by year, the corresponding NZ mean, the national range, and how Palmerston North City compares to the New Zealand average.

Indicator (Year)	TA Mean	NZ Mean	NZ Range	TA vs NZ Avg
PM2.5 (2018)	7.17	7.12	5.17-13.12	Similar
PM2.5 (2019)	7.24	6.71	4.96-10.47	Higher
PM2.5 (2020)	7.0	6.96	5.26-11.21	Similar
PM2.5 (2021)	5.38	5.52	4.12-10.34	Similar
PM2.5 (2022)	6.86	6.65	4.89-10.41	Higher

Table 2.3: Air Quality Indicator PM_{2.5} (µg/m³) - Palmerston North City vs National Averages

3 Housing Quality: Map & Population Share

This section examines the physical condition of housing across **Palmerston North City**. The indicators cover basic amenities, dampness, severe mould, and heating sources. All measures relate to **occupied private dwellings**, and higher percentages generally indicate poorer housing quality.

In 2023, 522 (1.8%) were always damp, 876 (3.0%) were severe mould, 84 (0.3%) were no basic amenities, and 363 (1.2%) were no heating source among occupied private dwellings. Overall, decreases for “Always damp”, “Severe mould”, “No basic amenities”, “No heating source” since 2018.

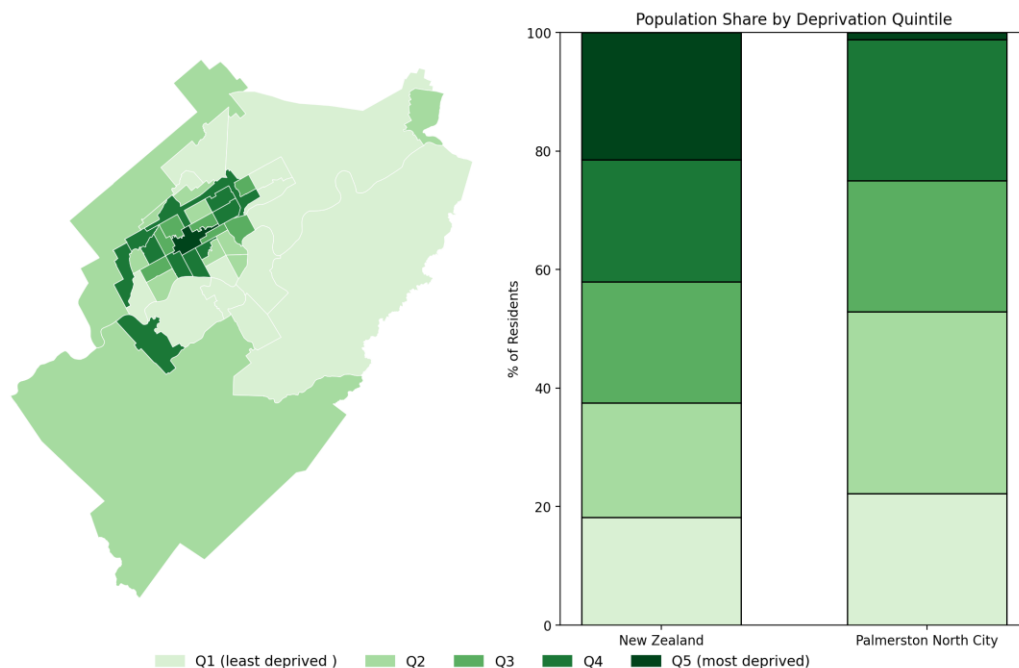


Figure 3: Housing Quality-Map & Population Share.

Figure 3 shows two simple views of housing conditions in **Palmerston North City**. The map on the left uses colours to show which neighbourhoods have more housing problems (darker colours) and which are doing better (lighter colours). The bar chart on the right shows how the people in **Palmerston North City** are spread across the five deprivation levels and compares this with the New Zealand average. This helps you see whether Palmerston North City has more people living in better-off areas or more people living in higher-need areas. Together, the map and chart show where housing challenges are located within the TA and how **Palmerston North City** compares with the rest of New Zealand overall.

3.1 Neighbourhood Patterns: Housing Quality

Figure 3.1 heatmap shows how housing conditions change as we move from the least deprived neighbourhoods in the TA to the most deprived ones. Neighbourhoods are first ranked using the composite score, which combines all indicators in the domain into a single overall measure of deprivation. These ranked areas are then split into groups, starting with Group 1 (least deprived) and ending with the last group (most deprived). Each row shows a different housing indicator, and each column shows the typical (median) value for that indicator within each deprivation group. Lighter colours mean better conditions, while darker colours mean worse conditions. If the colours get darker to the right, housing problems worsen with deprivation. If colours stay the same, the problems are evenly spread. If colours jump up and down, the problem is local to certain neighbourhoods, not linked to deprivation.

Note: When an SA2 has very few households, even a small change in the numerator can produce a large percentage, which may cause a darker cell in the heatmap. These patterns are normal in small-area data and should be interpreted as local variation rather than widespread issues across the TA.

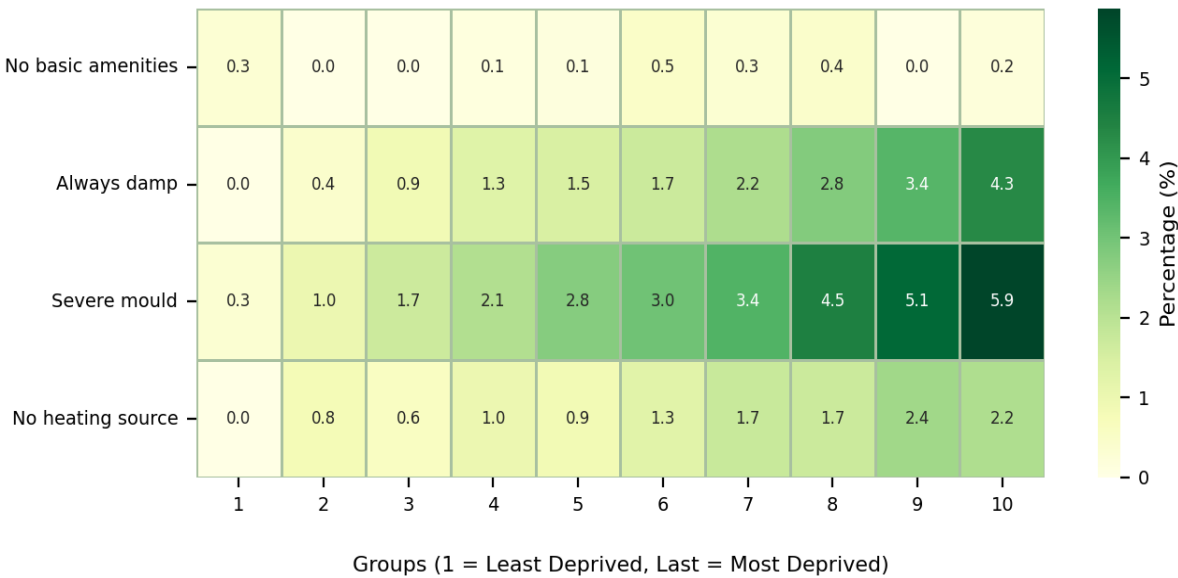


Figure 3.1: Housing Quality-Indicator Patterns Across Areas

3.2 Indicator Distribution Across Years: Housing Quality

Figure 3.2 compares how housing indicators are distributed across SA2 areas for the two most recent Census years. Each boxplot shows the typical value in the middle, the range where most neighbourhoods fall, and any unusually high values that appear as dots above the main box. A **higher** box position on the chart indicates that the indicator is generally worse or more common across the TA, while a **taller** box shows greater variation between neighbourhoods. By comparing the two years, it is possible to see whether conditions have improved, declined, or become more uneven over time. In smaller TAs, where the number of SA2s is limited, natural fluctuations between areas can appear more pronounced and should be interpreted with care.

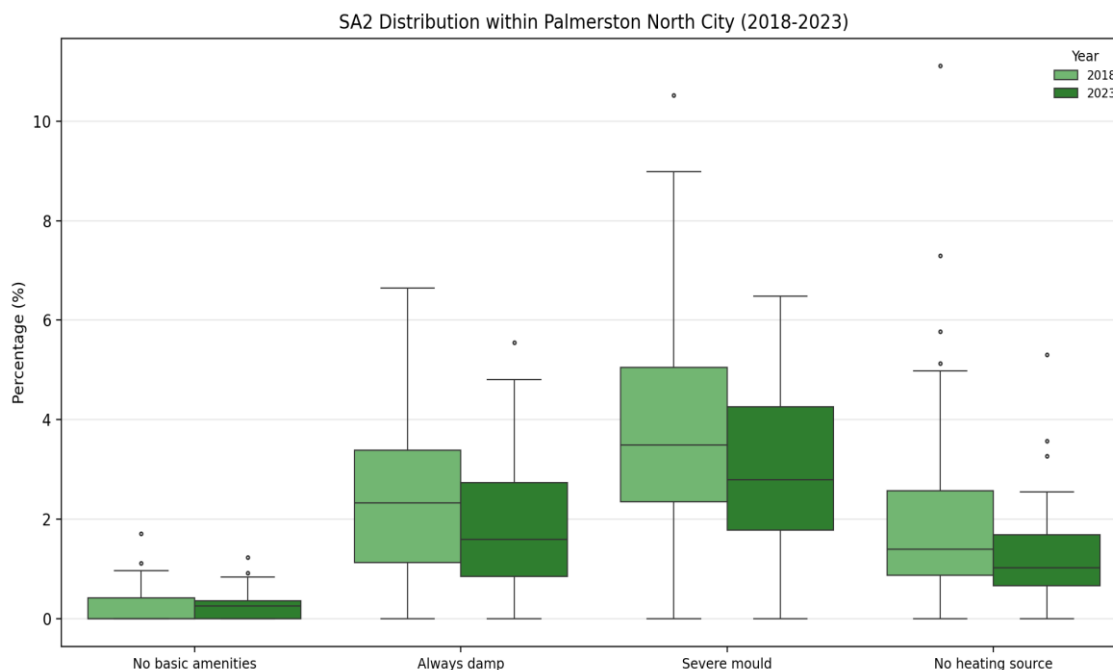


Figure 3.2: Housing Quality - Indicator Distribution Across Years

3.3 Comparison with National Average: Housing Quality

Figure 3.3 compares **Palmerston North City**'s results with national averages for each housing indicator. The solid circles represent the TA's values, while the dashed vertical lines mark the New Zealand average. The light green band shows the typical national range observed across all TAs. Indicators positioned to the right of the dashed line are worse than the national average, whereas those to the left indicate better outcomes. This comparison helps highlight whether housing conditions in **Palmerston North City** are broadly aligned with, better than, or worse than national patterns.

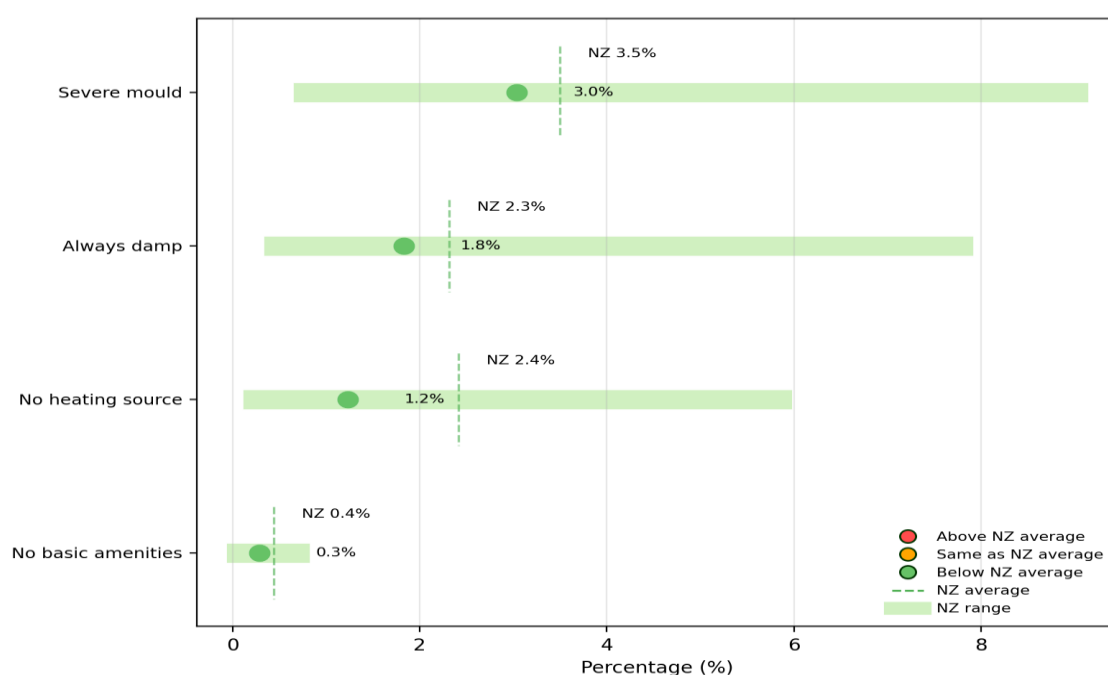


Figure 3.3: Housing Quality: Palmerston North City Comparison with National Levels.

4 Living Situation: Map & Population Share

This section describes the living conditions of households in occupied private dwellings. The indicators cover household crowding, access to motor vehicles, and access to telecommunication services. Higher values generally point to greater housing pressure or reduced access to transport and digital services.

In 2023, 1,515 (4.9%) were crowded households, 300 (1.0%) were no telecommunication, and 1,998 (6.8%) were no motor vehicle among households in occupied private dwellings. Overall, increases for “Crowded households”; while decreases for “No telecommunication”, “No motor vehicle” since 2018.

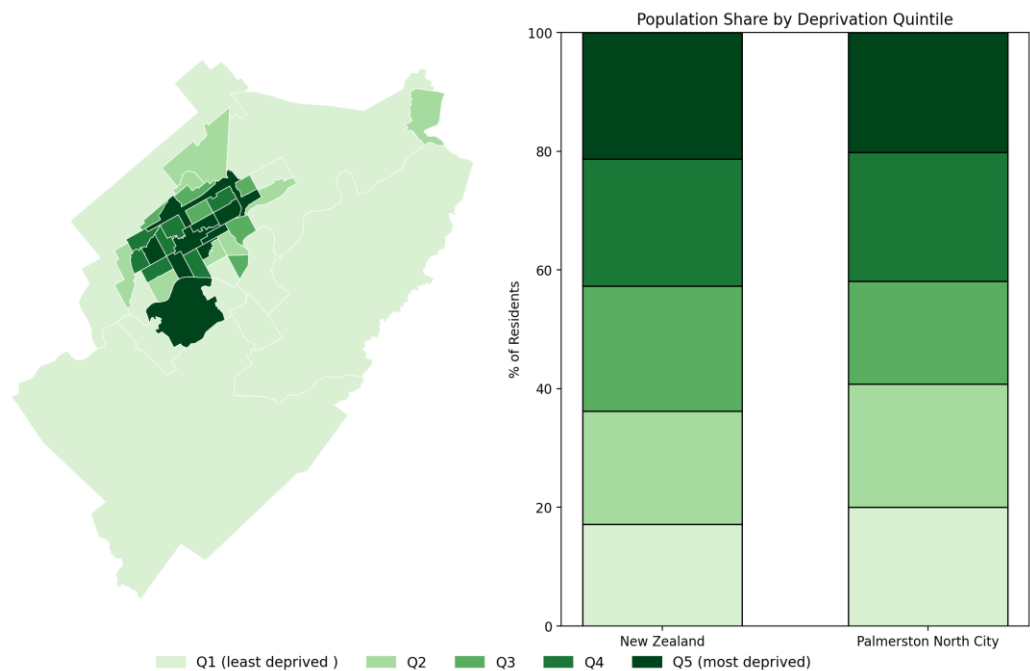


Figure 4: Living Situation- Map & Population Share

Figure 4 map shows how Living Situation conditions vary across SA2 areas in **Palmerston North City**. Darker areas represent neighbourhoods with higher levels of crowding, limited vehicle access, or no telecommunication. The bar chart compares **Palmerston North City** distribution with national patterns.

4.1 Neighbourhood Patterns: Living Situation

Figure 4.1 heatmap shows how Living Situation indicators change from lower-decile to higher-decile neighbourhoods. Darker shading represents higher percentages. Indicators with steady increases across deciles highlight stronger challenges in household space, connectivity, or transport.

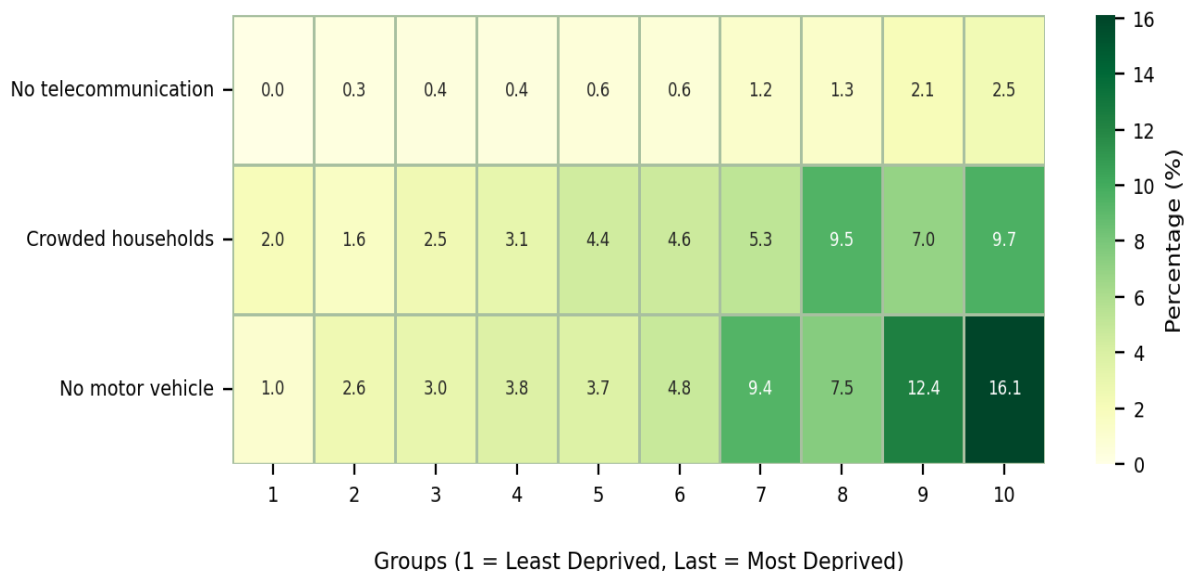


Figure 4.1: Living Situation -Indicator Patterns Across Areas.

4.2 Indicator Distribution Across Years: Living Situation

Figure 4.2 compares the distribution of crowding, vehicle access, and telecommunication indicators across SA2 areas for the two most recent Census years. The central line represents the typical (median) value, the box shows where most neighbourhoods sit, and dots above the box indicate unusually high areas. A higher box position suggests the issue is more common across the TA, while a taller box reflects greater differences between neighbourhoods. Differences may appear more pronounced in smaller TAs where the number of SA2s is limited.

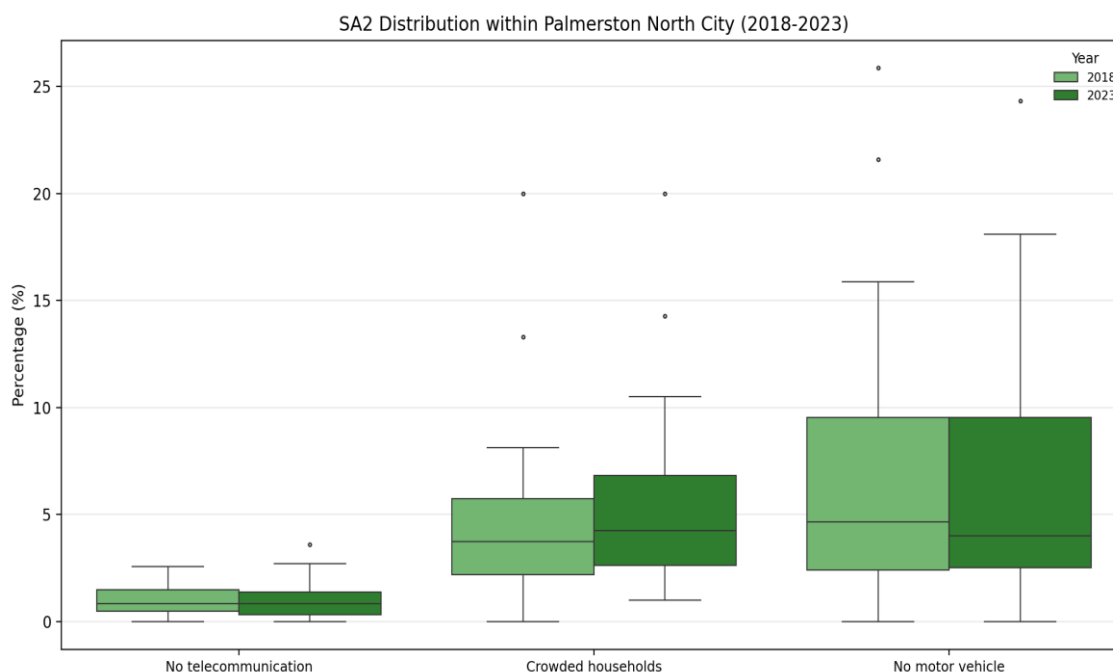


Figure 4.2: Living Situation- Indicator Distribution Across Years.

4.3 Comparison with National Average: Living Situation

Figure 4.3 compares Palmerston North City Living Situation indicators with national values. Circles represent local results, dashed lines mark the NZ average, and green bands show the usual NZ range.

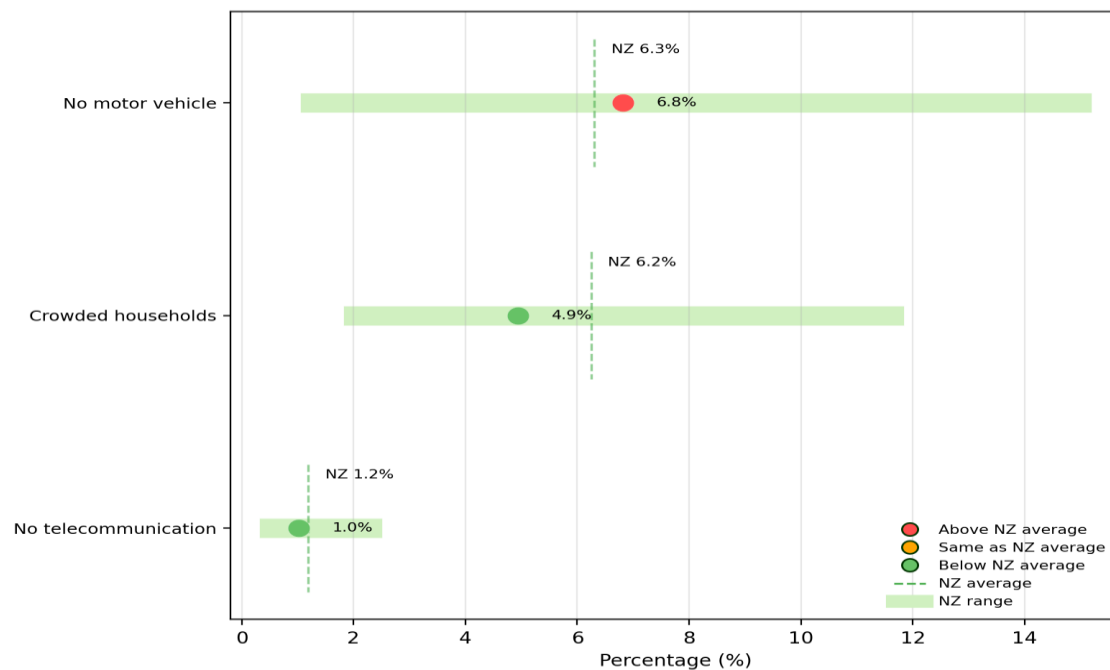


Figure 4.3: Living Situation - Palmerston North City Comparison with National Levels

5 Socioeconomic: Map & Population Share

This section explores socioeconomic wellbeing among the usually resident population aged 15 years and over. The indicators cover educational attainment, unemployment, reliance on Jobseeker and Sole Parent Support, and regular smoking rates. Higher values indicate greater socioeconomic hardship and inequality.

In 2023, 4,101 (5.8%) were jobseeker support, 10,968 (16.1%) were no qualification, 5,244 (7.5%) were regular smokers, 1,755 (2.5%) were sole parent support, and 2,160 (3.1%) were unemployment among individuals aged 15 years and over. Overall, increases for “Sole Parent Support”; while decreases for “Jobseeker Support”, “No qualification”, “Regular smokers”, “Unemployment” since 2018.

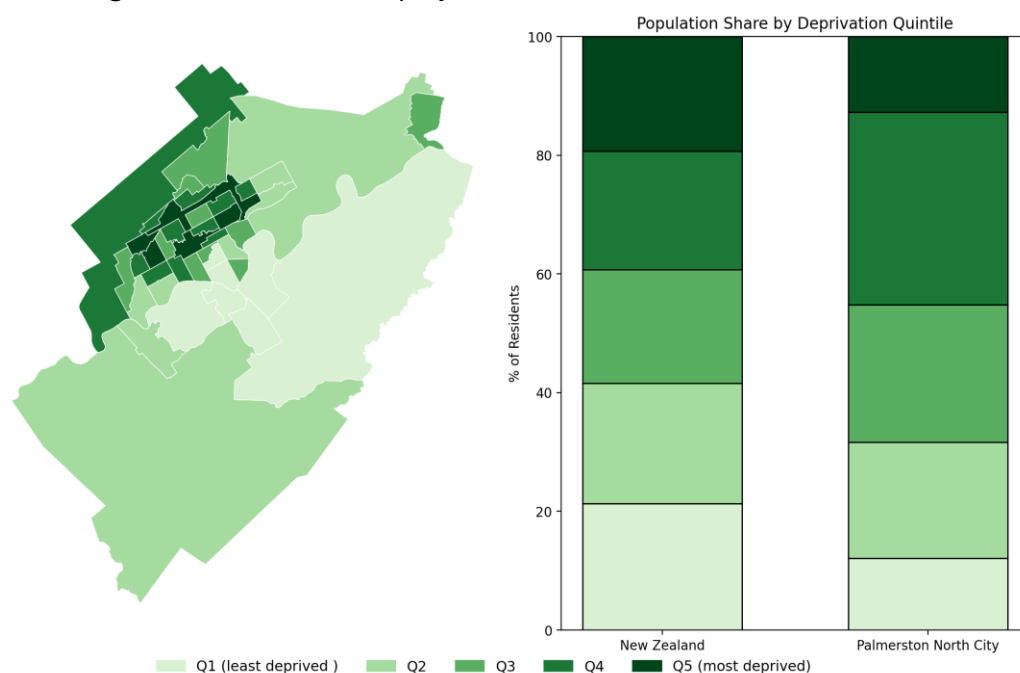


Figure 5: Socioeconomic - Map & Population Share

Figure 5 map shows how socioeconomic conditions differ across SA2 areas in Palmerston North City. Darker areas represent higher rates of unemployment, benefit reliance, lower qualifications, or regular smoking. The bar chart compares Palmerston North City with New Zealand overall.

5.1 Neighbourhood Patterns: Socioeconomic

Figure 5.1 heatmap shows how socioeconomic indicators change from low-decile to high-decile areas. Darker colours indicate higher levels of unemployment, fewer qualifications, greater benefit reliance, or higher smoking rates.

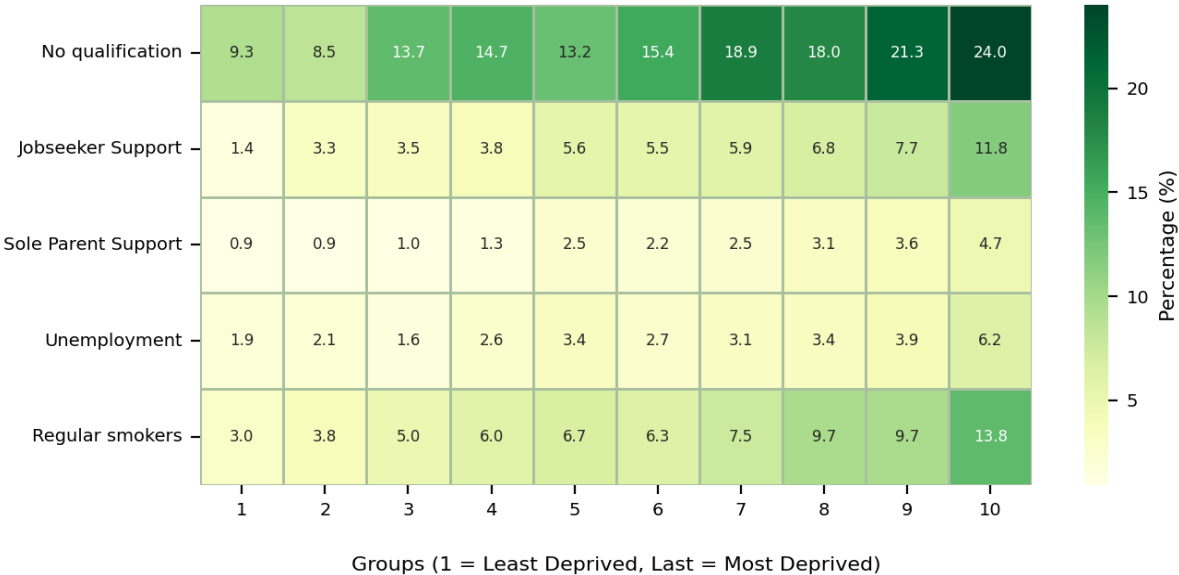


Figure 5.1: Socioeconomic-Indicator Patterns Across Areas

5.2 Indicator Distribution Across Years: Socioeconomic

Figure 5.2 chart compares the spread of socioeconomic indicators between the latest two Census years. Higher boxes indicate more common or more severe issues, while taller boxes show wider gaps across neighbourhoods.

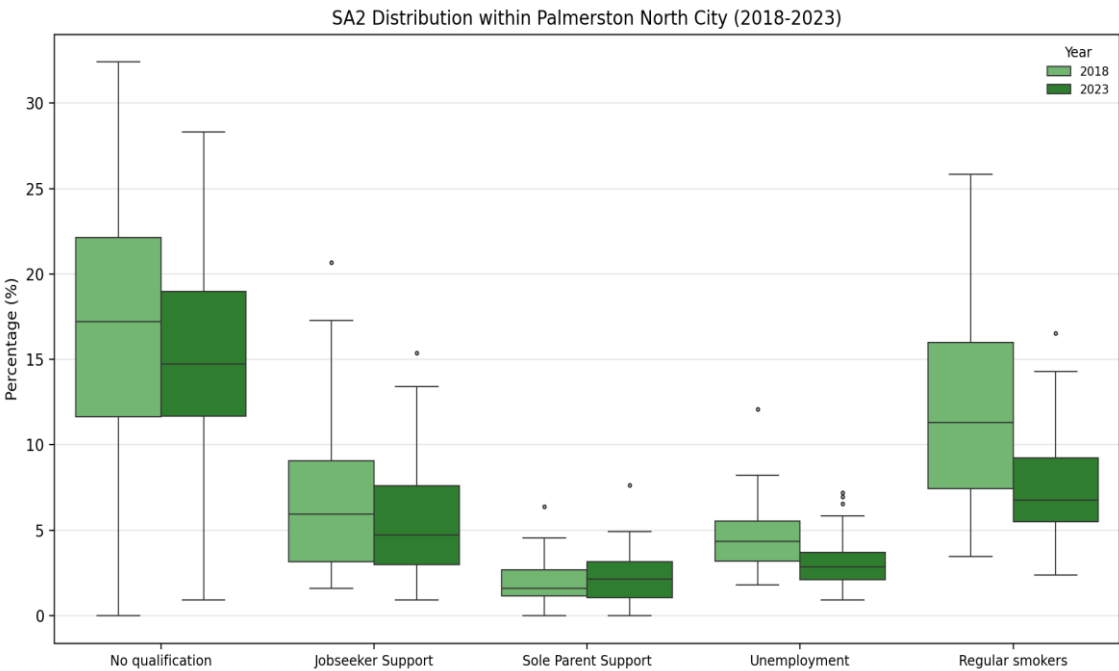


Figure 5.2: Socioeconomic-Indicator Distribution Across Years

5.3 Comparison with National Average: Socioeconomic

Figure 5.3 shows how Palmerston North City compares with national benchmarks for education, employment, and income support indicators. Points represent local values, and dashed lines show the NZ average.

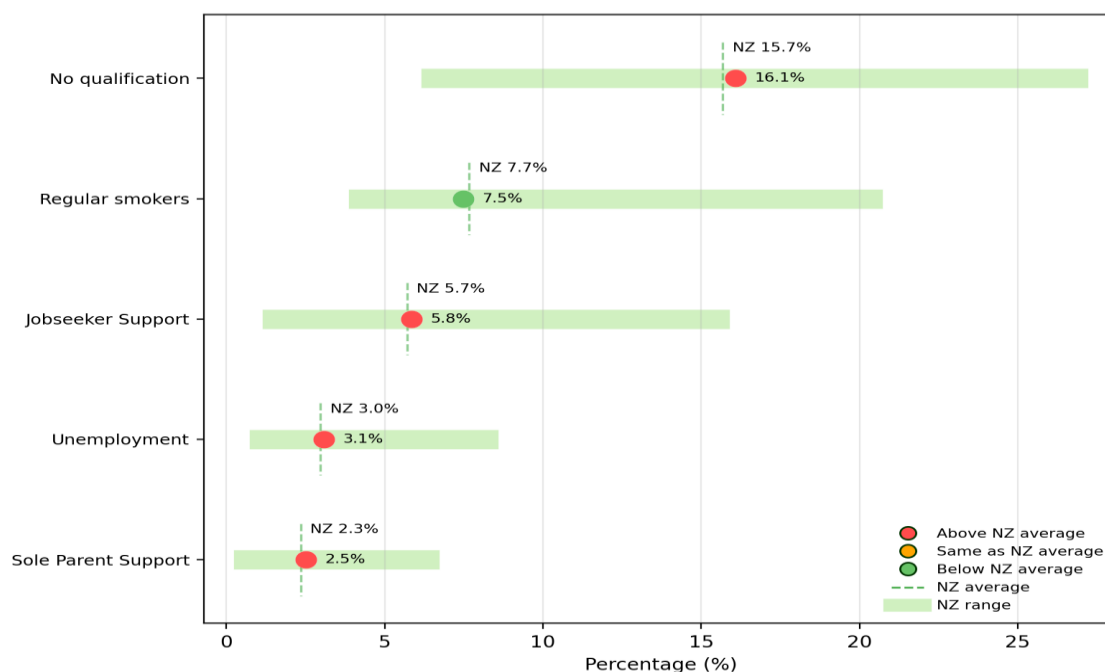


Figure 5.3: Socioeconomic- Palmerston North City Comparison with National Levels

6 Indicators Summary Table

This section brings together all key indicators from the Housing Quality, Living Situation, and Socioeconomic domains. Table 6 summarises Palmerston North City's local results and compares them with national averages and quartile ranges, providing a simple overview of where the area sits relative to the rest of New Zealand.

Indicator	TA Count	TA (%)	NZ Avg (%)	NZ LQ (%)	NZ UQ (%)	TA vs NZ Avg
No basic amenities	84	0.28	0.44	0.26	0.42	Lower
Always damp	522	1.83	2.31	1.39	2.67	Lower
Severe mould	876	3.04	3.5	2.11	4.11	Lower
No heating source	363	1.23	2.42	0.65	1.51	Lower
No telecommunication	300	1.02	1.19	0.93	1.45	Lower
Crowded households	1,515	4.95	6.25	3.13	5.92	Lower
No motor vehicle	1,998	6.81	6.31	3.73	6.17	Higher
No qualification	10,968	16.09	15.68	16.79	23.37	Higher
Jobseeker Support	4,101	5.84	5.72	4.29	6.93	Higher
Sole Parent Support	1,755	2.5	2.34	1.7	2.99	Higher

Unemployment	2,160	3.08	2.97	1.9	3.12	Higher
Regular smokers	5,244	7.47	7.67	7.69	11.14	Lower

Table 6: Summary of Key Indicators Compared with National Averages

7 Other Community Measures: Palmerston North City

This section presents a set of additional indicators describing rental housing patterns, unpaid work, family structure, and health-related difficulties. These indicators provide further insight into the social and support needs of residents across Palmerston North City.

7.1 Landlords

The landlord indicators show the distribution of rented households within **Palmerston North City** by landlord type. Figure 7.1 bar chart presents the number of households renting from Kāinga Ora, local councils, iwi or Māori trusts, and community housing providers for **2023**.

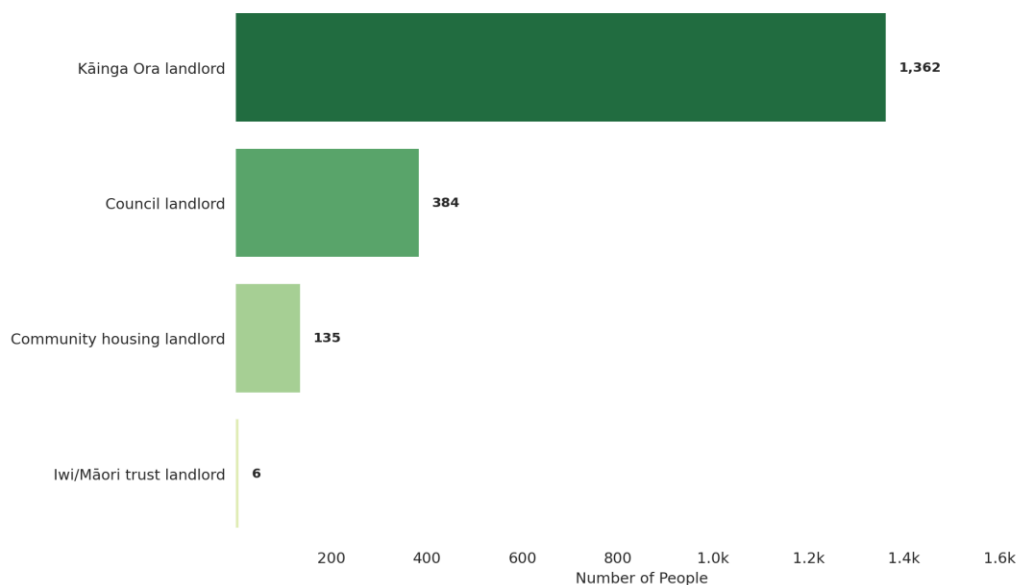


Figure 7.1: Landlord Types (2023)

Table 7.1 lists the local proportions alongside national values, allowing comparison between **Palmerston North City** and New Zealand overall.

Indicator	TA Count	TA (%)	NZ Avg (%)	NZ LQ (%)	NZ UQ (%)	TA vs NZ Avg
Council landlord	384	3.59	1.69	1.39	3.48	Higher
Kāinga Ora landlord	1,362	12.73	11.33	3.15	9.93	Higher
Iwi/Māori trust landlord	6	0.06	0.26	0.04	0.71	Lower
Community housing landlord	135	1.26	1.47	0.76	1.58	Lower

Table 7.1: Landlord Characteristics - Palmerston North City Compared with National Values

7.2 Unpaid Care and Support

The unpaid activities indicators describe the forms of unpaid work carried out by residents aged 15 and over in **Palmerston North City**. Figure 7.2 bar chart shows the number of people involved in caring for their own children, supporting ill or disabled family members, assisting people outside their household, and undertaking voluntary work during **2023**.

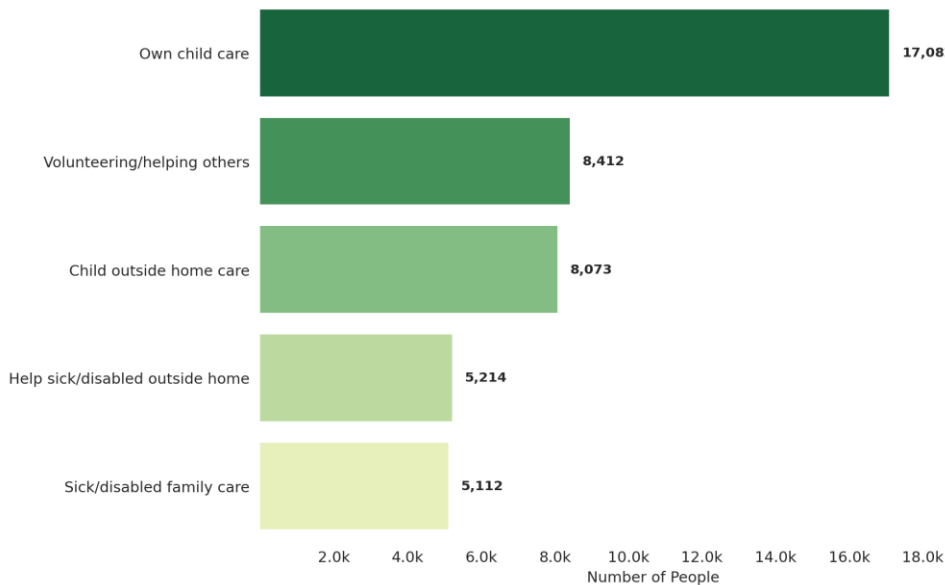


Figure 7.2: Unpaid Care and Support (2023)

Table 7.2 compares these local percentages with national benchmarks to show how **Palmerston North City** aligns with wider patterns of unpaid support.

Indicator	TA Count	TA (%)	NZ Avg (%)	NZ LQ (%)	NZ UQ (%)	TA vs NZ Avg
Own child care	17,082	28.39	28.39	25.42	29.3	Higher
Sick/disabled family care	5,112	8.5	7.85	7.49	9.07	Higher
Child outside home care	8,073	13.42	12.28	12.89	15.05	Higher
Help sick/disabled outside home	5,214	8.67	7.86	8.15	9.86	Higher
Volunteering/helping others	8,412	13.98	13.32	14.02	18.55	Higher

Table 7.2: Unpaid Care and Support Indicators - Palmerston North City vs National Results

7.3 Family Situation

The family situation indicators describe the relationship status of people aged 15 and older in **Palmerston North City**. Figure 7.3 bar chart illustrates the number of residents who are married, separated, divorced, widowed, or never partnered in **2023**.

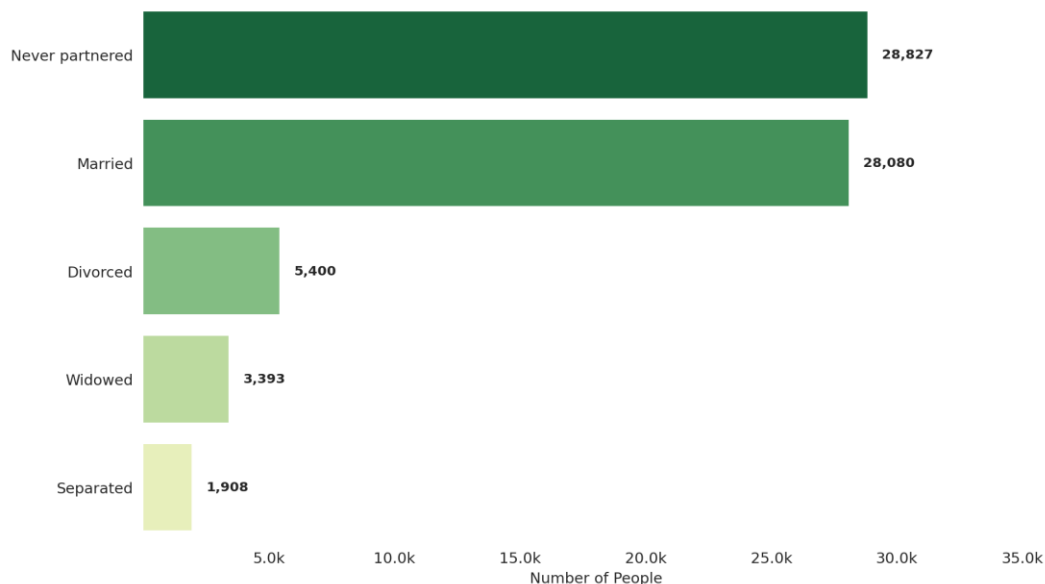


Figure 7.3: Family Situation (2023)

Table 7.3 provides local values alongside national averages, offering context for how relationship patterns in **Palmerston North City** compare with the rest of New Zealand.

Indicator	TA Count	TA (%)	NZ Avg (%)	NZ LQ (%)	NZ UQ (%)	TA vs NZ Avg
Married	28,080	41.52	44.74	42.15	48.48	Lower
Separated	1,908	2.82	2.77	2.81	3.29	Higher
Divorced	5,400	7.98	7.92	7.99	9.78	Higher
Widowed	3,393	5.02	4.94	5.26	6.58	Higher
Never partnered	28,827	42.62	39.57	32.81	40.73	Higher

Table 7.3: Family Situation Indicators - Palmerston North City vs National Results

7.4 Disability and Health Difficulties

The health and disability indicators show the number of people aged 5 and over in **Palmerston North City** who experience difficulty with communication, hearing, vision, mobility, memory, or self-care. Figure 7.4 bar chart summarises the counts for each type of difficulty for **2023**.

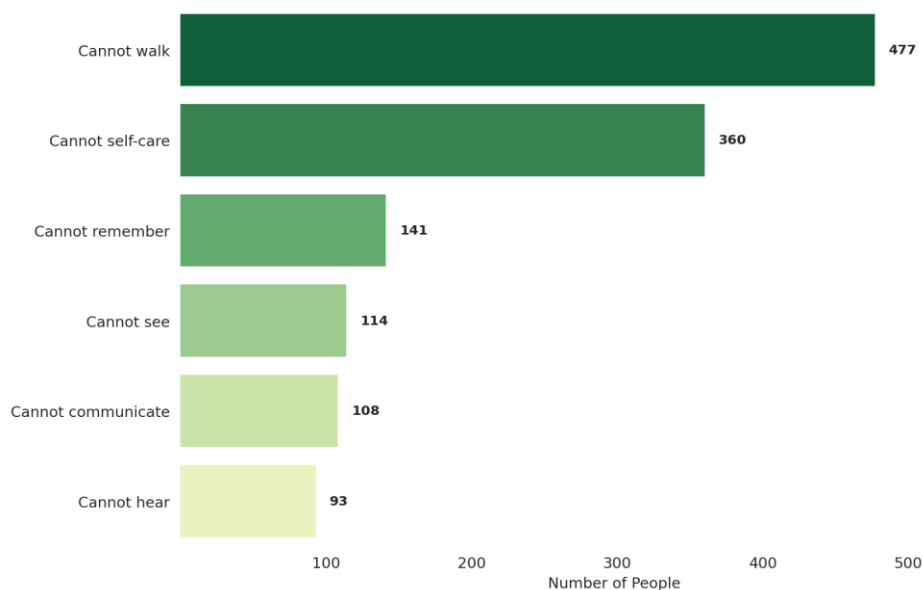


Figure 7.4: Health and Disability Difficulties (2023)

Table 7.4 includes local percentages and national reference values, showing the relative frequency of these health-related limitations within **Palmerston North City**.

Indicator	TA Count	TA (%)	NZ Avg (%)	NZ LQ (%)	NZ UQ (%)	TA vs NZ Avg
Cannot communicate	108	0.15	0.16	0.11	0.19	Lower
Cannot hear	93	0.13	0.15	0.09	0.16	Lower
Cannot remember	141	0.2	0.21	0.17	0.26	Lower
Cannot see	114	0.16	0.18	0.13	0.2	Lower
Cannot walk	477	0.67	0.49	0.41	0.62	Higher
Cannot self-care	360	0.51	0.42	0.33	0.48	Higher

Table 7.4: Health and Disability Difficulties - Palmerston North City Compared with National Values